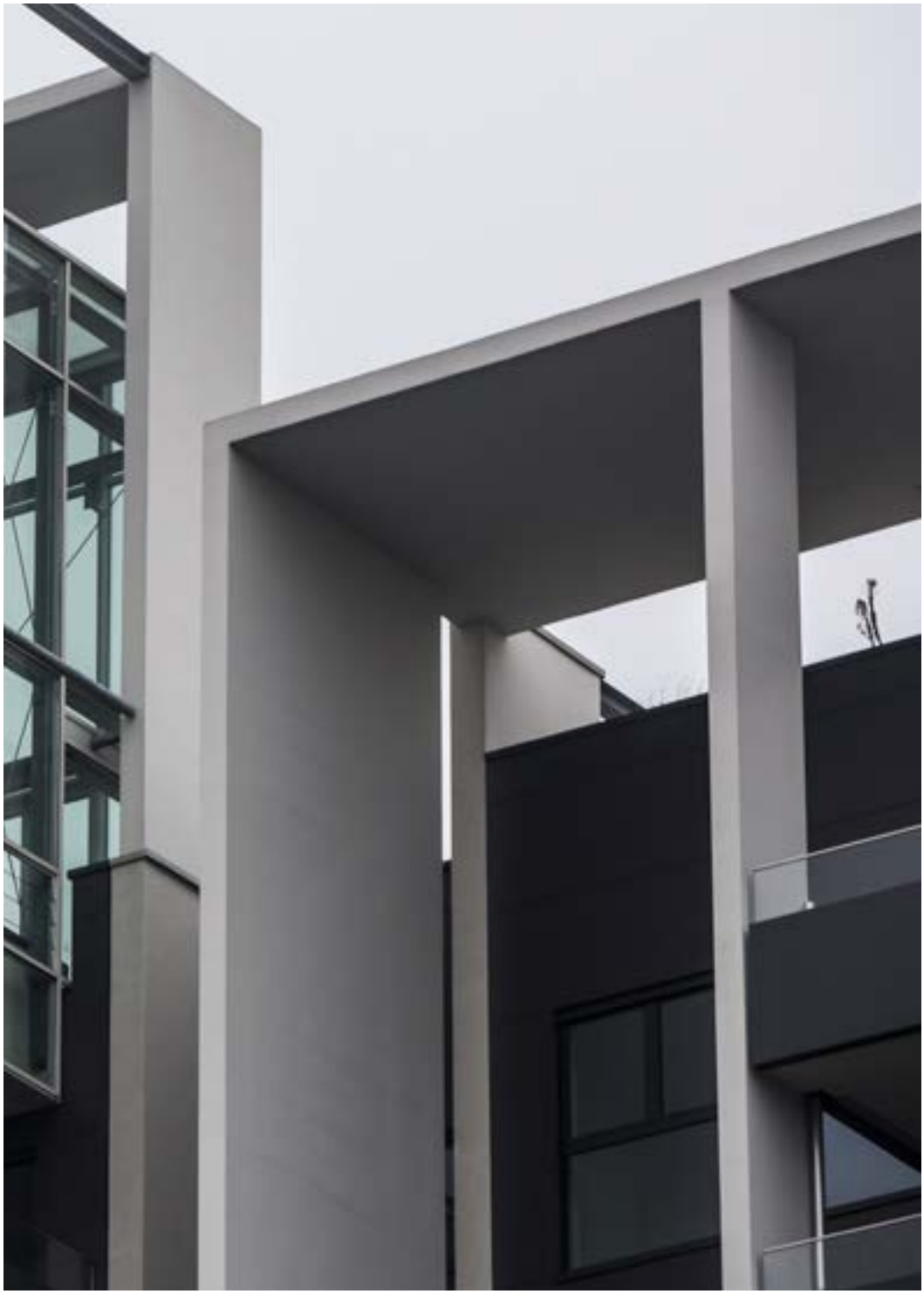
Nexion

ITALIAN STYLE THAT INSPIRES

Ventilated facades



Nexion
ARCHITECTURE DIVISION

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Ventilated facades

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Nexion Architecture Division

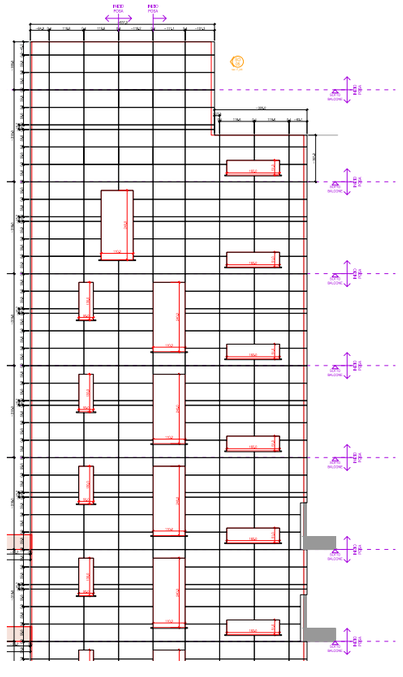
THE BEST ITALIAN STYLE FOR MAJOR ARCHITECTURAL PROJECTS

Nexion is a brand created by Emilgroup, an Italian firm that is amongst the world's leading manufacturers of ceramic surfaces, and Simpolo Vitrified, one of the top Indian ceramics and sanitary fitting producers.

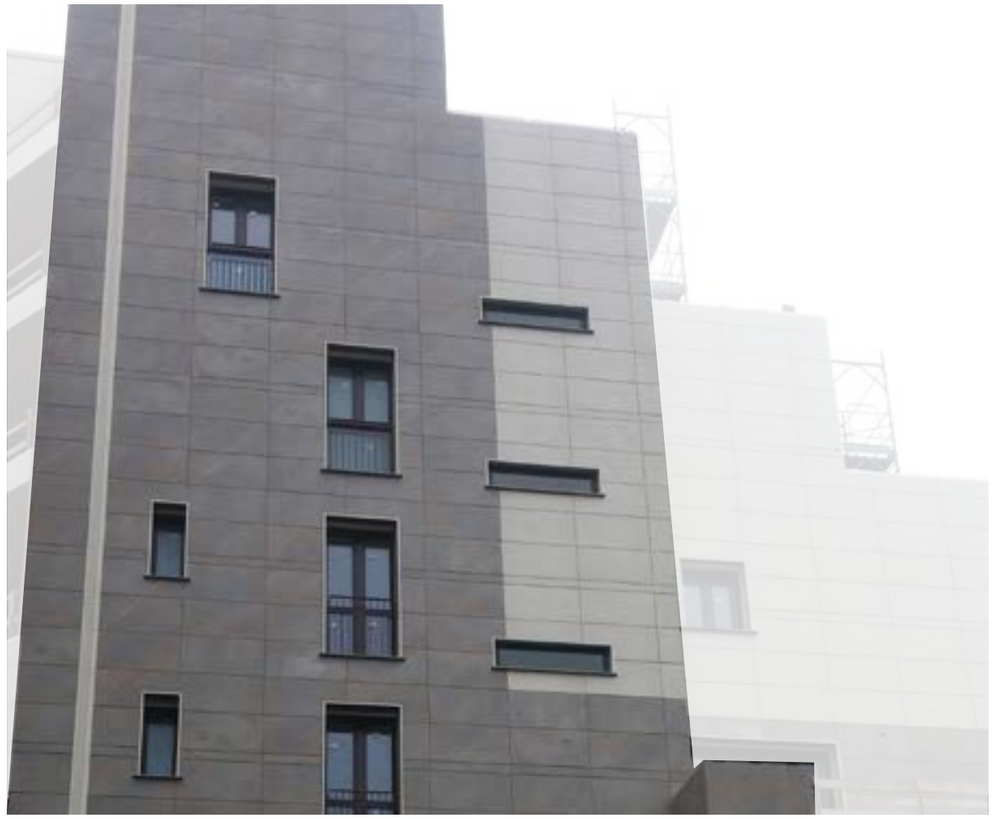
Nexion creates a new bridge between Italy and India, by offering ceramic surfaces and design solutions of impeccable quality, designed entirely in Italy and produced in India using the very latest Italian technologies.

With Architecture Division, Nexion assumes the role of complete consultant, capable of assisting the customer from the very earliest design phases to installation and certification, and also introduces the very best Italian style - in terms of appearance, technical properties and product quality - into the field of major architectural projects, by producing surface covering systems for indoor and outdoor use, such as ventilated facades, raised floorings and partition walls.

With its tailor-made approach, Nexion Architecture Division assures projects' exclusiveness, safety and quality, as well as the maximum control of all factors, both internal (building design and layout) and external (positioning of the building in relation to wind direction, seismic risk, nearness to the sea or extreme altitudes).



Design



Realisation

Nexion Design and Performance

TWO MANUFACTURERS OF EXCELLENCE TO ENSURE BUILDINGS' STYLISTIC AND PHYSICAL QUALITY

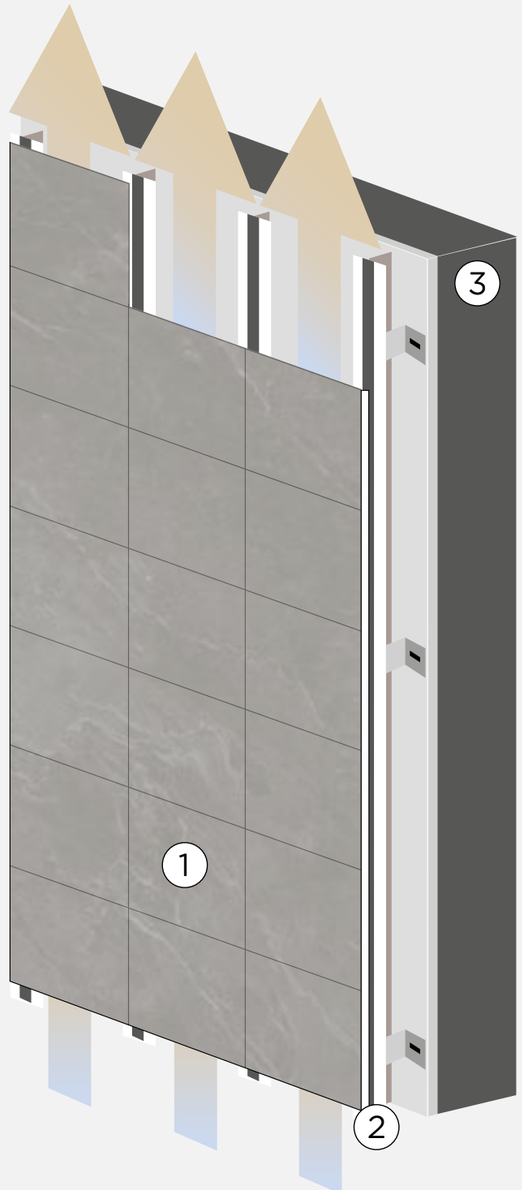
Ventilated facades are created by cladding the outsides of buildings with ceramic materials and they provide excellent results in terms of both appearance and buildings insulation against high and low temperature, noise and weathering.

All this is possible thanks to the "stack effect", a rising flow of air which ensures the natural ventilation of the external walls. The effect is produced by forming a cavity between the external cladding layer and the walls of the building, filled with thermal and acustic insulation material.

Nexion ventilated facades are engineered by Terramilano, leader in the design and construction of this type of solution both in Italy and abroad, with a 25 years experience in this field.

A partnership that is an inspired blend of leading-edge technical expertise and excellent construction capabilities: Terramilano provides the skeleton, by building structures of extremely high quality and performance, while Nexion provides ceramic materials of peerless appearance and technical and functional characteristics.

Stack effect



1 Nexion surfaces
2 Ventilated facades structure
3 Building structure

The benefits of ventilated facades

DESIGN, CONVENIENCE AND ENERGY SAVING: AN UNBEATABLE SOLUTION

DESIGN
Nexion ventilated facades allow buildings to be finished with ceramic materials of the finest quality and exquisite appearance.

ENERGY SAVING
The thermal insulation provided by ventilated facades considerably reduces the costs of air-conditioning in summer and heating in winter.

INTERNAL COMFORT
The insulating jacket created by a ventilated facades raises the comfort level inside buildings all year round, by keeping the temperature constant in both summer and winter.

QUICK, ECONOMICAL INSTALLATION
For both new buildings and renovation projects, ventilated facades, can be installed quickly and at reasonable costs. Maintenance operations are also safe, quick and economical.

DURABILITY OVER TIME
The very high quality of the materials used to create the bearing structure and the ventilated facades construction method guarantee excellent performance in terms of durability, with additional positive returns in terms of sustainability.

SOUNDPROOFING
Ventilated facades significantly reduce noise levels, cutting them by up to one half.

ELIMINATION OF COLD BRIDGES
Combined with a suitable layer of insulating material, ventilated facades create a uniform insulating jacket around buildings, eliminating thermal bridges and minimizing temperature variations.

VENTILATION
The ventilation generated through the cavity between the outer cladding and the insulating panel provides a natural temperature regulation to the buildings. The “stack effect” forms an effective thermal shield that disperses excess heat in summer and provides protection against the cold in winter.

PROTECTION FROM DAMP AND ACCIDENTAL INFILTRATIONS
The ventilation generated removes the risk of condensate, maintaining the perfect conditions of the internal structure.

MAXIMUM COMPATIBILITY
Ventilated facades can be installed on buildings of all kinds, both continuous and discontinuous, and are compatible with ceramic of any size, colour, design or thickness.

NEW BUILDINGS AND RENOVATION PROJECTS
Ventilated facades can be constructed both to clad new buildings and to upgrade existing ones by increasing their energy efficiency.

FEWER STRESSES APPLIED TO BUILDINGS
Ventilated facades considerably reduce the weight of cladding materials applied to buildings and reduce the risk of detaching, failure or breakage, thanks to the lower weight of the materials used and the specially designed anchor fixtures.

The benefits of Nexion surfaces

STYLE, BEAUTY AND STRENGTH

INSPIRED BY ITALIAN STYLE
Nexion surfaces express the very best of contemporary Italian design.

DURABLE BEAUTY
Prolonged exposure to the sun does not cause any fading in the decoration or colours of Nexion surfaces.

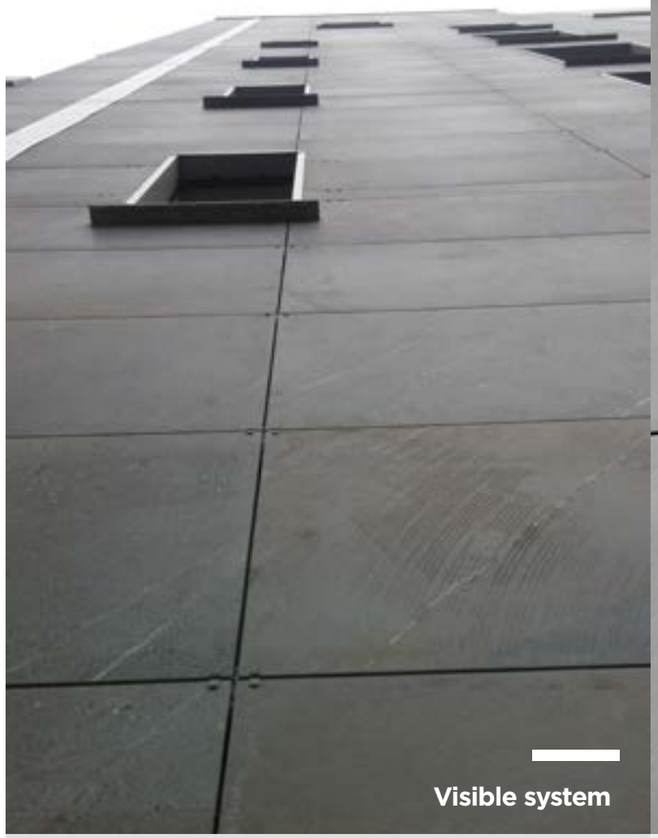
OUTSTANDING RESISTANCE
Nexion surfaces are extremely resistant to chemicals, smog and the thermal shocks triggered by phenomena such as long heatwaves or sudden rainfall that causes sharp falls in outdoor temperatures.



Ventilated facades anchor systems

VISIBLE, HIDDEN AND BONDED:
THREE SYSTEMS FOR ALL ARCHITECTURAL NEEDS





Visible system



Hidden system

Bonded system



Nexion ventilated facades engineered by Terramilano are designed to cover all the various needs of engineers, architects and designers in terms of appearance, function and structure. This is achieved thanks to three specific anchor systems, allowing the realisation of different types of solutions.

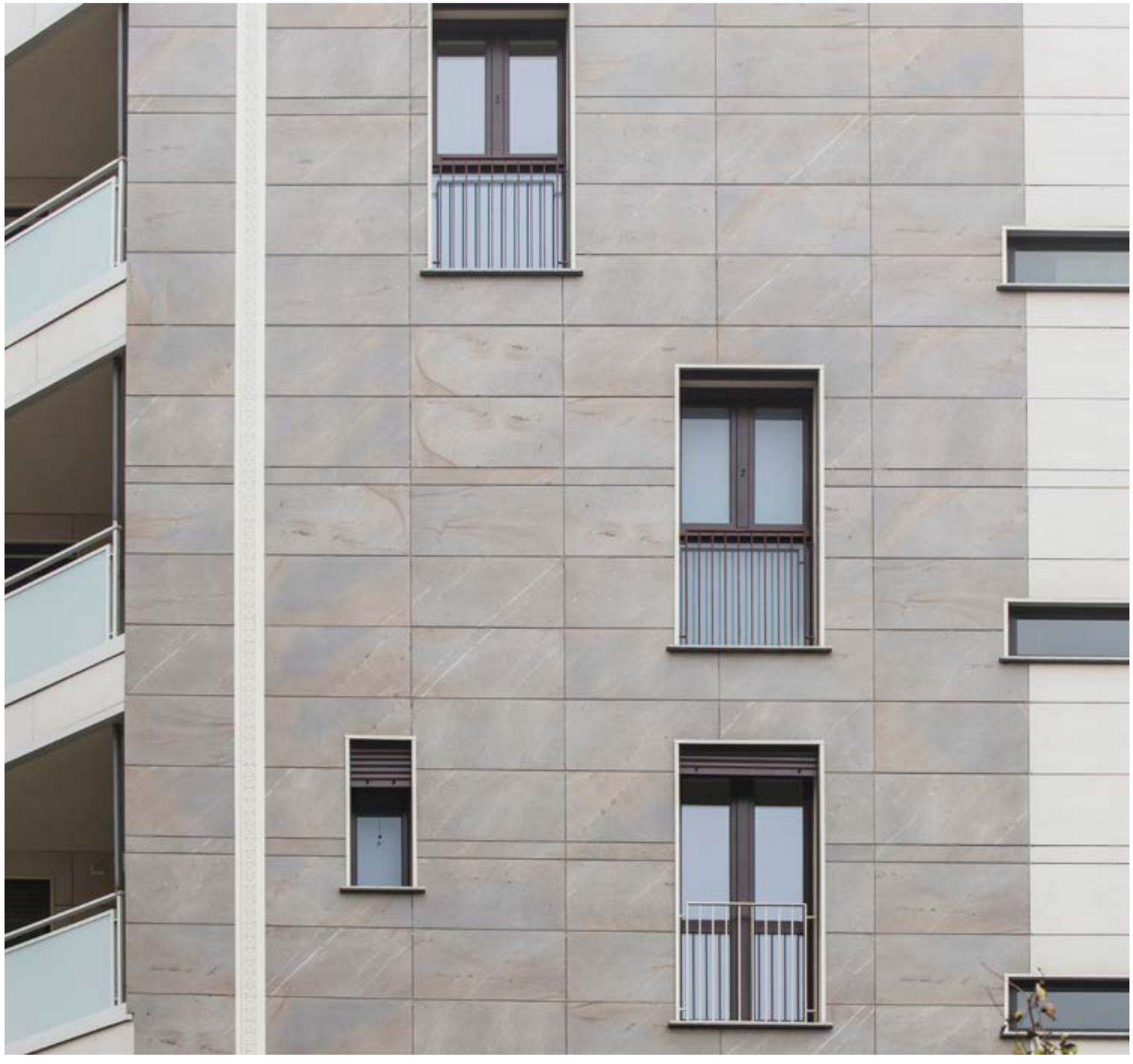
Visible system

With this solution, the system used to anchor the cladding material to the structure is visible on the outside.

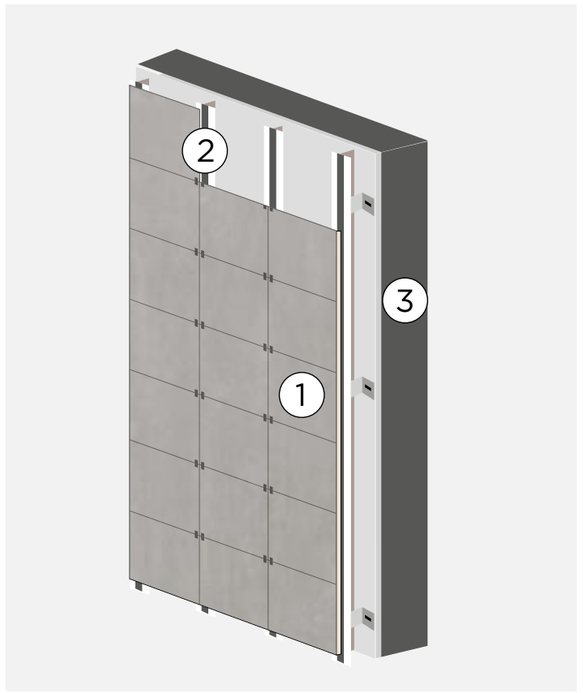
The slabs are held in place by clips fixed to the structure with drilling screws or blind rivets.

For a less visible anchor system, the clips can be painted in RAL colours to match the slabs.





Visible system



1 Nexion surfaces
2 Visible system
3 Building structure



1 Vertical stand TRUST80®
2 Bracket ST80®
3 Bolt stainless steel
4 Hook A.M.4
5 Neoprene
6 Nexion slab

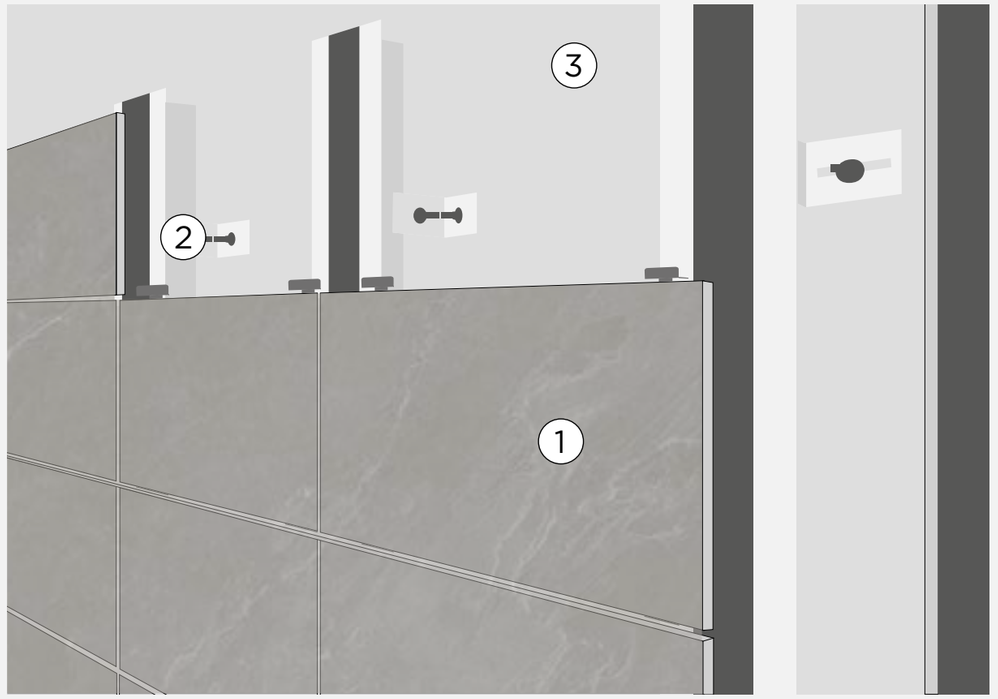
Hidden system

With this solution, the system used to anchor the cladding material to the structure is not visible on the outside.

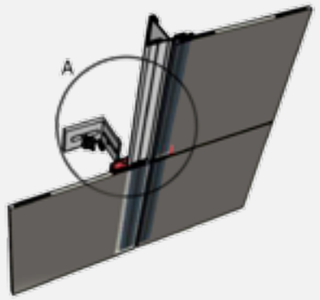
This gives a more seamless appearance. Before installation, the slabs are cut to accommodate the anchor fixtures.

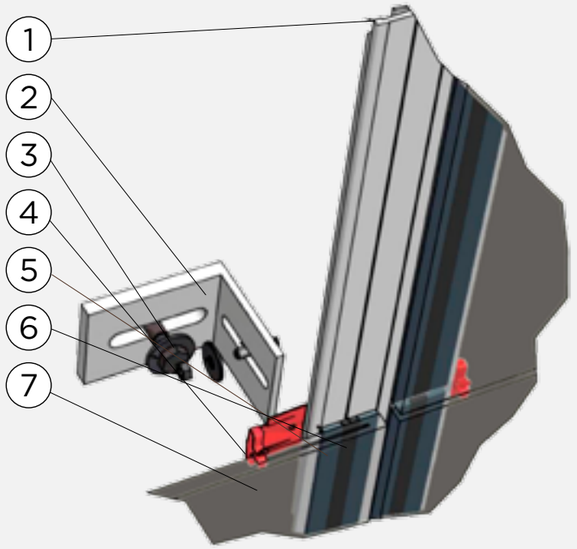


Hidden system



1 Nexion surfaces
2 Hidden system
3 Building structure





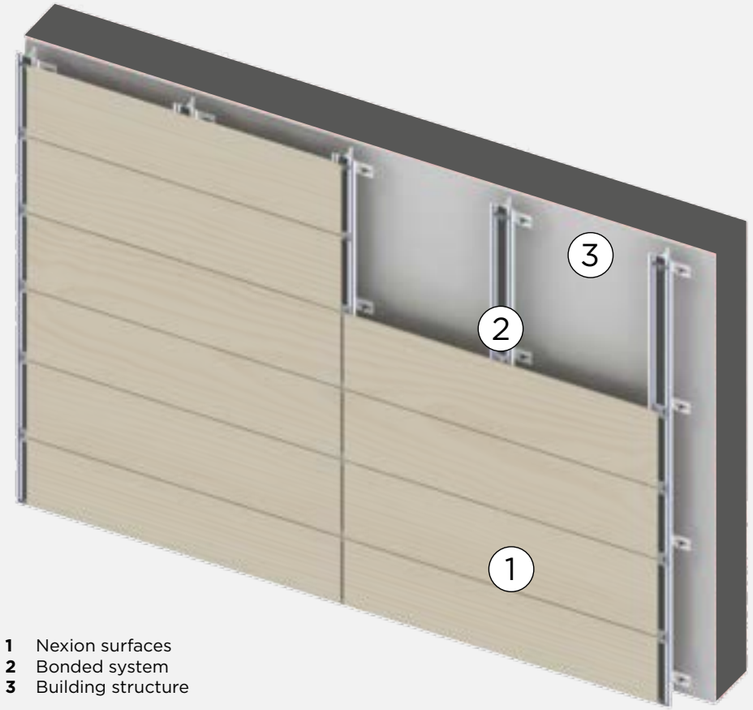
1 Vertical stand TRUST80®
2 Bracket ST80®
3 Mechanical chemical anchor fastener
4 S10®
5 Structural adhesive
6 Neoprene
7 Nexion slab

Bonded system

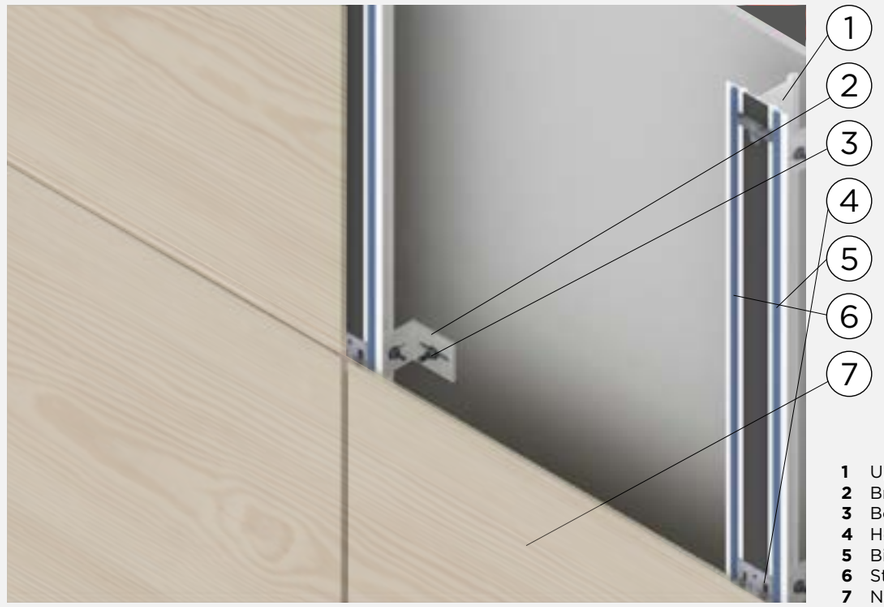
In this system, the cladding materials are anchored in place with structural adhesives, combined with safety hooks which support the weight of the slabs and provide very high levels of resistance.

This system is recommended for nexion slabs' large size and slimline thickness. It allows irregular installation layouts with irregular alignment.





1 Nexion surfaces
2 Bonded system
3 Building structure



1 Upright TRUST80®
2 Bracket ST80®
3 Bolt stainless steel
4 Hook
5 Biadhesive safety tape
6 Structural adhesive
7 Nexion slab

Technical info

Nexion collections

Endless

thickness **9 mm**
thickness **6 mm** 120x240 cm


NERO


GRIGIO

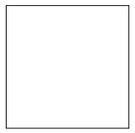

BIANCO


AVORIO


BEIGE


TAUPE


120x240 cm
F14ENCK0NR0 Nero
F14ENCG0NR0 Grigio
F14ENCA0NR0 Bianco
F14ENCB1NR0 Avorio
F14ENCB0NR0 Beige
F14ENCF0NR0 Taupe


120x120 cm
F12END0NR0 Nero
F12END0NR0 Grigio
F12ENDA0NR0 Bianco
F12END1NR0 Avorio
F12END0NR0 Beige
F12ENDF0NR0 Taupe


80x160 cm
F86END0NR0 Nero
F86END0NR0 Grigio
F86ENDA0NR0 Bianco
F86END1NR0 Avorio
F86END0NR0 Beige
F86ENDF0NR0 Taupe


80x80 cm
F80END0NR0 Nero
F80END0NR0 Grigio
F80ENDA0NR0 Bianco
F80END1NR0 Avorio
F80END0NR0 Beige
F80ENDF0NR0 Taupe


60x120 cm
F62END0NR0 Nero
F62END0NR0 Grigio
F62ENDA0NR0 Bianco
F62END1NR0 Avorio
F62END0NR0 Beige
F62ENDF0NR0 Taupe


60x60 cm
F60END0NR0 Nero
F60END0NR0 Grigio
F60ENDA0NR0 Bianco
F60END1NR0 Avorio
F60END0NR0 Beige
F60ENDF0NR0 Taupe


30x60 cm
F63END0NR0 Nero
F63END0NR0 Grigio
F63ENDA0NR0 Bianco
F63END1NR0 Avorio
F63END0NR0 Beige
F63ENDF0NR0 Taupe

Coniwood

thickness **9 mm**


ABETE TINTO


LARICE OSSIDATO


LARICE


ABETE


PINO

CEDRO SBIANCATO


30x240 cm
F34CNWA0NR0 Cedro Sbiancato
F34CNWB0NR0 Pino
F34CNWB1NR0 Abete
F34CNWC0NR0 Larice
F34CNWG0NR0 Larice Ossidato
F34CNWK0NR0 Abete Tinto


20x240 cm
F24CNWA0NR0 Cedro Sbiancato
F24CNWB0NR0 Pino
F24CNWB1NR0 Abete
F24CNWC0NR0 Larice
F24CNWG0NR0 Larice Ossidato
F24CNWK0NR0 Abete Tinto


30x160 cm
F36CNWA0NR0 Cedro Sbiancato
F36CNWB0NR0 Pino
F36CNWB1NR0 Abete
F36CNWC0NR0 Larice
F36CNWG0NR0 Larice Ossidato
F36CNWK0NR0 Abete Tinto


20x160 cm
F26CNWA0NR0 Cedro Sbiancato
F26CNWB0NR0 Pino
F26CNWB1NR0 Abete
F26CNWC0NR0 Larice
F26CNWG0NR0 Larice Ossidato
F26CNWK0NR0 Abete Tinto


20x120 cm
F22CNWA0NR0 Cedro Sbiancato
F22CNWB0NR0 Pino
F22CNWB1NR0 Abete
F22CNWC0NR0 Larice
F22CNWG0NR0 Larice Ossidato
F22CNWK0NR0 Abete Tinto

Runa

thickness **9 mm**
thickness **6 mm** 120x240 cm


GRIGIO CHIARO


GRIGIO SCURO


BEIGE


NERO


120x240 cm
F14RNAG0NR0 Grigio Chiaro
F14RNAB0NR0 Beige
F14RNAG1NR0 Grigio Scuro
F14RNAK0NR0 Nero

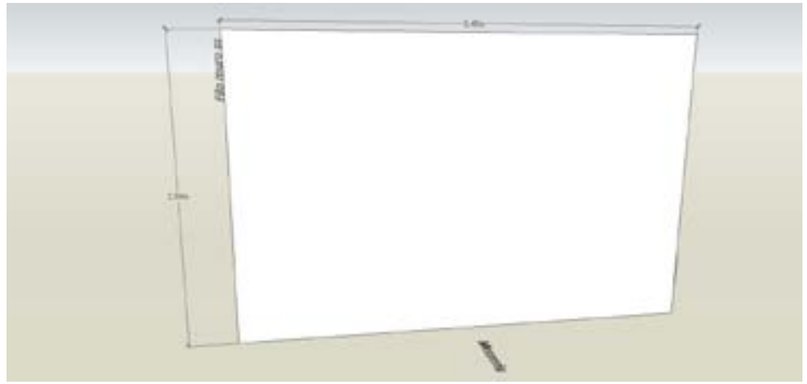

80x160 cm
F86RNAG0NR0 Grigio Chiaro
F86RNAB0NR0 Beige
F86RNAG1NR0 Grigio Scuro
F86RNAK0NR0 Nero


60x120 cm
F62RNAG0NR0 Grigio Chiaro
F62RNAB0NR0 Beige
F62RNAG1NR0 Grigio Scuro
F62RNAK0NR0 Nero


60x60 cm
F60RNAG0NR0 Grigio Chiaro
F60RNAB0NR0 Beige
F60RNAG1NR0 Grigio Scuro
F60RNAK0NR0 Nero

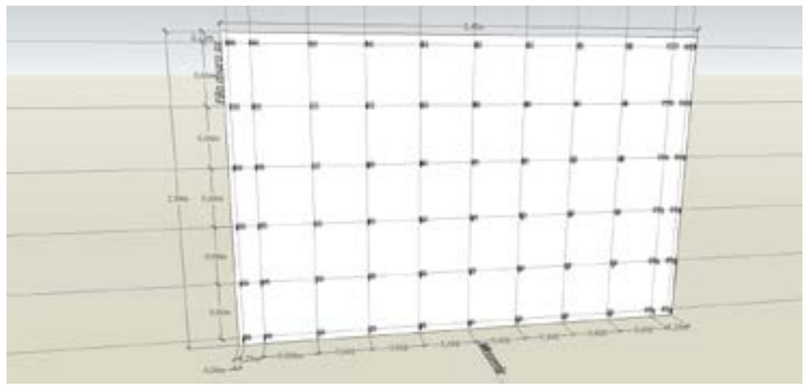

30x60 cm
F63RNAG0NR0 Grigio Chiaro
F63RNAB0NR0 Beige
F63RNAG1NR0 Grigio Scuro
F63RNAK0NR0 Nero

Structure installation phases

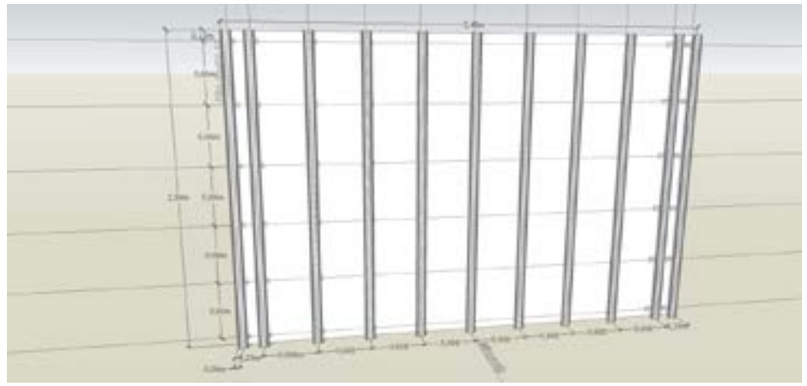


1 Bare wall

2 Marking out



3 Drilling of holes and fitting of brackets



4 Fitting of vertical stand

5 Installation of Nexion surfaces compatible with the chosen anchor system

Certifications and calculation reports

Nexion delivers a full consulting package from the initial design of the ventilated facades through to final delivery.

Every project is certified and guaranteed through technical drawings, calculation reports and documentation containing the specific design characteristics.


ERICA

RAPPORTO DI PROVA No. 1124

TERRAMILANO S.r.l.
Via Don Verzeri, 18
20152 - Milano
ITALIA

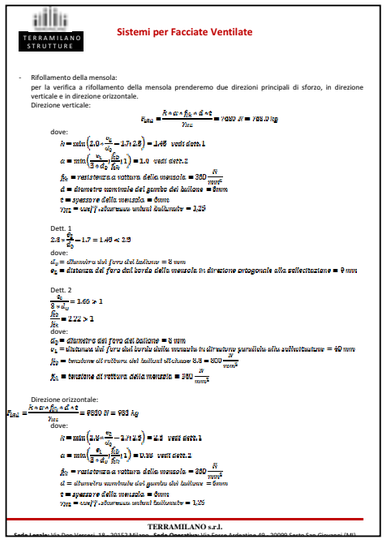
E.R.I.C.A. SOC. COOP. A R.L.
LABORATORIO TECNOLOGICO PROVE DI QUALITÀ SU MATERIALI LAPIDEI E MATERIALI COMPOSITI

PROVE ESEGUITE:

1. Prove di Trazione meccaniche

Il Rapporto di Prova No. 1124 composto da 8 pagine inclusa questa.

Anchor system test report


TERRAMILANO STRUTTURE

Sistemi per Facciate Ventilate

- Rifollamento della mensola:
per la verifica a rifollamento della mensola prenderemo due direzioni principali di sforzo, in direzione verticale e in direzione orizzontale.
Direzione verticale:

$$\sigma_{Rd} = \frac{R + \alpha \cdot f_{bv} \cdot d \cdot s}{\gamma_{M2}} = 9320\ N = 950.9\ kg$$

dove:
$k = \min\left(2.0 - \frac{e_2}{d_0} = 2.7;2.5\right) = 2.5$ vedi sect.1
$\alpha = \min\left(\frac{e_1}{3 \cdot d_0}; \frac{f_{bv}}{f_{bv}} = 1\right) = 1.0$ vedi sect.2
f_{bv} = resistenza a rottura della mensola = 360 $\frac{N}{mm^2}$
d = diametro nominale del gambo del bullone = 8mm
t = spessore della mensola = 6mm
γ_{M2} = coeff. riduzione valori bullonato = 1.25

Dett. 1
$2.8 \cdot \frac{e_2}{d_0} - 1.7 = 1.46 < 2.5$
dove:
d_0 = diametro del foro del bullone = 9 mm
e_2 = distanza del foro del bordo della mensola in direzione ortogonale alla sollecitazione = 9 mm

Dett. 2
$\frac{e_1}{3 \cdot d_0} = 1.66 > 1$
$\frac{f_{ub}}{f_{bv}} = 2.22 > 1$
dove:
d_0 = diametro del foro del bullone = 9 mm
e_1 = distanza del foro del bordo della mensola in direzione parallela alla sollecitazione = 40 mm
f_{ub} = tensione di rottura del bullone sollecitato 8.8 = 800 $\frac{N}{mm^2}$
f_{bv} = tensione di rottura della mensola = 360 $\frac{N}{mm^2}$

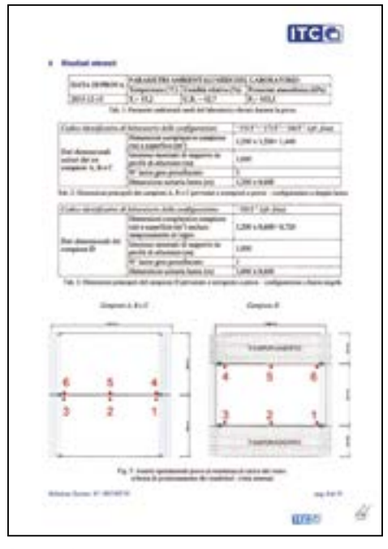
Direzione orizzontale:

$$F_{Rd} = \frac{k \cdot \alpha \cdot f_{bv} \cdot d \cdot s}{\gamma_{M2}} = 9320\ N = 950\ kg$$

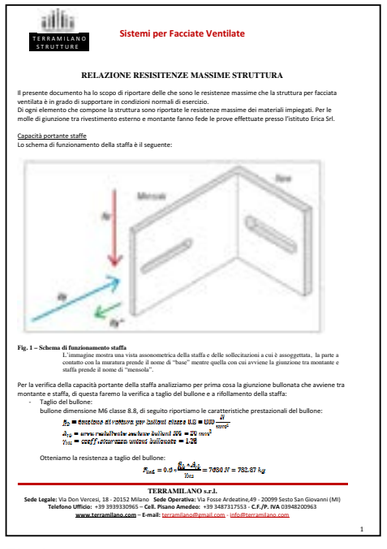
dove:
$k = \min\left(2.8 \cdot \frac{e_2}{d_0} - 1.7 = 2.5;2.5\right) = 2.5$ vedi sect.1
$\alpha = \min\left(\frac{e_1}{3 \cdot d_0}; \frac{f_{ub}}{f_{bv}} = 1\right) = 0.92$ vedi sect.2
f_{bv} = resistenza a rottura della mensola = 360 $\frac{N}{mm^2}$
d = diametro nominale del gambo del bullone = 8mm
t = spessore della mensola = 6mm
γ_{M2} = coeff. riduzione valori bullonato = 1.25

TERRAMILANO s.r.l.

Stress calculations


ITCQ

Strength test


TERRAMILANO STRUTTURE

Sistemi per Facciate Ventilate

RELAZIONE RESISITENZE MASSIME STRUTTURA

Il presente documento ha lo scopo di riportare delle che sono le resistenze massime che la struttura per facciata ventilata è in grado di supportare in condizioni normali di esercizio.
Di ogni elemento che compone la struttura sono riportate le resistenze massime dei materiali impiegati. Per le molle di giunzione tra rivestimento esterno e montante fanno fede le prove effettuate presso l'istituto Erica Srl.

Capacità portante staffe
Lo schema di funzionamento della staffa è il seguente:

Fig. 1 – Schema di funzionamento staffa
L'immagine mostra una vista assonometrica della staffa e delle sollecitazioni a cui è assoggettata, la parte a contatto con la muratura prende il nome di "base" mentre quella con cui avviene la giunzione tra montante e staffa prende il nome di "mensola".

Per la verifica della capacità portante della staffa analizziamo per prima cosa la giunzione bullonata che avviene tra montante e staffa, di questa faremo la verifica a taglio del bullone e a rifollamento della staffa:
- Taglio del bullone:
bullone dimensione M6 classe 8.8, di seguito riportiamo le caratteristiche prestazionali del bullone:
f_u = resistenza a rottura per bulloni classe 8.8 = 800 $\frac{N}{mm^2}$
A_{VS} = area resistente a taglio bullone M6 = 20 mm^2
γ_{M2} = coeff. riduzione valori bullonato = 1.25

Ottenuto la resistenza a taglio del bullone:

$$F_{Rd} = 0.6 \cdot \frac{f_u \cdot A_{VS}}{\gamma_{M2}} = 7680\ N = 782.87\ kg$$

TERRAMILANO s.r.l.
Sede Legale: Via Don Verzeri, 18 - 20152 Milano **Sede Operativa:** Via Fosse Ardeatine,49 - 20099 Sesto San Giovanni (MI)
Telefono Ufficio: +39 3939330965 – **Cell. Pisano Amedeo:** +39 3487317553 - **C.F./P. IVA** 03948200963
www.terramilano.com – E-mail: terramilano@gmail.com; info@terramilano.com

1

Final report

ISO 13006: Annex G - EN 14411: Annex G. Dry - Pressed Ceramic Tiles.

	TECHNICAL FEATURE	STANDARD METHOD ISO	REQUIREMENTS		NEXION Values			
					Side < 1200 mm		Side ≥ 1200 mm	
			%	mm	%	mm	%	mm
DIMENSIONS AND SURFACE QUALITY	LENGH AND WIDTH	UNI EN ISO 10545-2	±0,6%	±2,0	±0,2%	±2,0	±0,1%	±1,5
	THICKNESS		±5%	±0,5	±5%	±0,5	±4%	±0,36
	STRAIGHTNESS OF SIDES		±0,5%	±1,5	±0,2%	±1,5	±0,1%	±1,5
	RECTANGULARITY		±0,5%	±2,0	±0,33%	±1,5	±0,1%	±1,5
	FLATNESS		±0,5%	±2,0	±0,26%	±1,5	±0,2%	±1,5
	SURFACE QUALITY		A minimum of 95% of the tiles shall be free from visible defects that would impair the appearance of a major area of tiles.		COMPLIANT			
PHISICAL PROPERTIES	WATER ABSORPTION	UNI EN ISO 10545-3	$E_b \leq 0,5\%$ **Individual maximum : 0,6%**		$E_b \leq 0,2\%$			
	BREAKING STRENGTH	UNI EN ISO 10545-4	Min 1300 N		≥ 2000 N			
	MODULUS OF RUPTURE		R ≥ 35 N/mm² minimum 32 Not applicable to tiles with breaking strength ≥ 3000 N		R ≥ 45 N/mm² minimum 40 Not applicable to tiles with breaking strength ≥ 3000 N			
	FROST RESISTANCE	UNI EN ISO 10545-12	REQUIRED		COMPLIANT			
	SCRATCH HARDNESS (Mohs scale)	EN 101	Min 6		COMPLIANT			
	SURFACE ABRASION	UNI EN ISO 10545-7	0 (100 cycles) 1 (150 cycles) 2 (600 cycles) 3 (1500 cycles) 4 (6000 cycles) 5 (> 12000 cycles)		5			
	COEFFICIENT OF LINEAR THERMAL EXPANSION FROM AMBIENT TEMPERATURE TO 100° C (K^{-1})	ISO 10545-8	REQUIRED		6 x 10^{-6} K^{-1} max			
	CRAZING RESISTANCE	ISO 10545-11	REQUIRED		COMPLIANT			
	RESISTANCE TO THERMAL SHOCK	ISO 10545-9	10 CYCLES Min		COMPLIANT			
	IMPACT RESISTANCE	ISO 10545-5	Min 0,55		COMPLIANT			
	BULK DENSITY	DIN 51082	2,2 g/cc Min		COMPLIANT			
CHEMICAL PROPERTIES	RESISTANCE TO STAINING	UNI EN ISO 10545-14	Minimum Class 3		5			
	RESISTANCE TO CHEMICALS	UNI EN ISO 10545-13	Minimum UB		GA			
	TO HOUSEHOLD CHEMICALS AND SWIMMING POOL SALTS							
	TO HIGH AND LOW CONCENTRANTIONS OF ACIDS AND ALKALIS		To be reported		GLA - GHA			
	SLIP RESISTANCE	DIN 51130			R10			
		DIN 51097			B (A+B)			

ALL THE TEST ARE CARRIED OUT BY ITALIAN LABORATORIES ACCREDITED ACCORDING TO ISO/IEC 17025:2005

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ITALIAN STYLE THAT INSPIRES

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